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```c
#include <stdlib.h>

#include <time.h>

#define MAX_FRAMES 50

void sender(int frames[], int size, int window_size);

void receiver(int frames[], int start, int window_size);

int main() {

    int n, window_size;

    int frames[MAX_FRAMES];

    printf("Sliding Window Protocol Simulation (Go-Back-N)\n");

    printf("Enter number of frames to send (max %d): ", MAX_FRAMES);

    scanf("%d", &n);

    if(n > MAX_FRAMES || n <= 0) {

        printf("Invalid number of frames.\n");

        return 0;

    }

    printf("Enter window size: ");

    scanf("%d", &window_size);

    if(window_size <= 0 || window_size > n) {

        printf("Invalid window size.\n");

        return 0;

    }

    // Fill frames with sequence numbers

    for(int i=0; i<n; i++) {

        frames[i] = i;

    }

    sender(frames, n, window_size);

    return 0;
```

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```c
void sender(int frames[], int size, int window_size) {

    int base = 0;

    int next_seq = 0;

    char choice;

    srand(time(NULL));

    while(base < size) {

        printf("\nSender window: ");

        for(int i=base; i<base+window_size && i<size; i++) {

            printf("[%d] ", frames[i]);

        }

        printf("\nSending frames...\n");

        for(int i=base; i<base+window_size && i<size; i++) {

            printf("Sent frame: %d\n", frames[i]);

        }

        printf("\nSimulating receiver...\n");

        receiver(frames, base, window_size);

        printf("\nDid all frames receive correctly? (y/n): ");

        scanf(" %c", &choice);

        if(choice == 'y' || choice == 'Y') {

            base += window_size;

        } else {

            printf("Resending from frame %d...\n", frames[base]);

            // In Go-Back-N, we resend from base

        }
```

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```
107            printf("Resending from frame %d...\n", frames[base]);
108
109            // In Go-Back-N, we resend from base
110
111        }
112
113    }
114
115    printf("\nAll frames sent successfully!\n");
116
117 }
118
119 void receiver(int frames[], int start, int window_size) {
120
121    for(int i=start; i<start+window_size; i++) {
122
123        printf("Received frame: %d, sending ACK: %d\n", frames[i],
124
125    }
126
127 }
```

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```
Sliding Window Protocol Simulation (Go-Back-N)
Enter number of frames to send (max 50): 5
Enter window size: 4

Sender window: [0] [1] [2] [3]
Sending frames...
Sent frame: 0
Sent frame: 1
Sent frame: 2
Sent frame: 3

Simulating receiver...
Received frame: 0, sending ACK: 1
Received frame: 1, sending ACK: 2
Received frame: 2, sending ACK: 3
Received frame: 3, sending ACK: 4

Did all frames receive correctly? (y/n): y

Sender window: [4]
Sending frames...
Sent frame: 4

Simulating receiver...
Received frame: 4, sending ACK: 5
Received frame: 24298, sending ACK: 24299
Received frame: 1265472352, sending ACK: 1265472353
Received frame: 30507, sending ACK: 30508

Did all frames receive correctly? (y/n): y

All frames sent successfully!


...Program finished with exit code 0
Press ENTER to exit console.
```